

REED PUTNAM PRACTICE · FALL 2026

Number Theory: Congruences, Orders & Valuations

Week 4 · big numbers, small moduli, one prime at a time

Thu Oct 15 / Fri Oct 16

The toolbox (1)

Orders

For $\gcd(a, n) = 1$, $\text{ord}_n(a)$ is the least $d \geq 1$ with $a^d \equiv 1 \pmod{n}$. Then $a^k \equiv 1 \iff d \mid k$; so $d \mid \varphi(n)$, and $d \mid p-1$ when $n = p$.

- > **Fermat / Euler:** $a^{p-1} \equiv 1 \pmod{p}$ for $p \nmid a$; $a^{\varphi(n)} \equiv 1 \pmod{n}$ for $\gcd(a, n) = 1$.
Reduce exponents mod $\varphi(n)$, **never mod n** .
- > **Corollaries:** $2^a - 1 \mid 2^b - 1 \iff a \mid b$; $q \mid a^p - 1, q \nmid a - 1 \Rightarrow q \equiv 1 \pmod{p}$.
- > **Wilson:** $(p-1)! \equiv -1 \pmod{p}$ — pair each k with its inverse.

The toolbox (2)

- > CRT: $\gcd(m, n) = 1 \Rightarrow \mathbb{Z}/mn \cong \mathbb{Z}/m \times \mathbb{Z}/n$. Work one prime power at a time, then reassemble.
- > Valuations: $v_p(ab) = v_p(a) + v_p(b)$; $v_p(a + b) \geq \min(v_p(a), v_p(b))$, equality when they differ.
- > Legendre: $v_p(n!) = \sum_{k \geq 1} \lfloor n/p^k \rfloor = \frac{n - s_p(n)}{p - 1}$.
- > Lifting the exponent: p odd, $p \mid x - y$, $p \nmid xy \Rightarrow v_p(x^n - y^n) = v_p(x - y) + v_p(n)$.
- > gcd tricks: coprime factors of a perfect power are perfect powers; $xy + \alpha x + \beta y = c$ becomes $(x + \beta)(y + \alpha) = c + \alpha\beta$: a divisor count.

Example 1

The last two digits

Find the last two digits of 7^{2026} .

→ work it on the board (~2 min)

Example 2

Wilson, symmetrized

For an odd prime p ,

$$\left(\frac{p-1}{2}\right)!^2 \equiv (-1)^{(p+1)/2} \pmod{p}.$$

→ work it on the board (~3 min)

Example 3

A valuation that cannot cancel

For $n \geq 2$, $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ is not an integer.

→ work it on the board (~3 min)

How to recognize it

- > Absurdly large numbers (7^{2026}): reduce, then compute an order.
- > “All n with $n \mid 2^n - 1$ ”, “every prime divisor of...”: name the order, squeeze.
- > Factorials, binomials, “never an integer”: a valuation, at 2 or the largest prime.
- > $1/a + 1/b = c/N$: factor, count divisors. $k \mapsto mk \bmod n$: CRT in disguise.

Pitfalls

- > Fermat needs $p \nmid a$, Euler needs $\gcd(a, n) = 1$: $2^{\varphi(4)} \not\equiv 1 \pmod{4}$.
- > Exponents reduce mod $\varphi(n)$ or the order; only bases reduce mod n .

Now you

Problems 1–4.

1–2 warm-ups · 3–4 easy Putnam · 5–6 take-home

35 minutes, groups of 2–3

Solutions posted after Friday.